Practice workbook · Explain how a blocked coronary artery damages the heart

Causes and consequences of narrowing or blocking arteries supplying heart muscle or other tissue.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

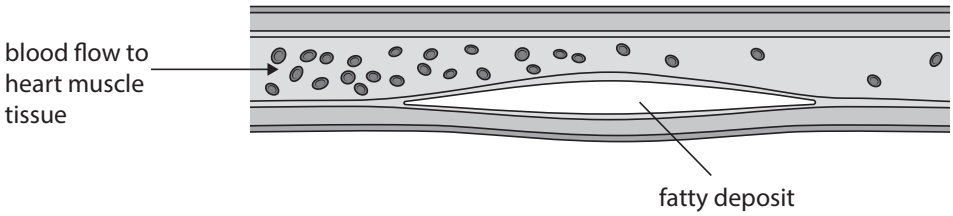
Type ID: HBI-EXPLAIN-CORONARY-BLOCKAGE-EFFECTS

1. Question 7 · part a · 4 marks

June 2024 Paper 2

Worked-example retry · Attempt part a only.

7 (a) The diagram shows a fatty deposit in an artery carrying blood to the heart muscle.



(Source: adapted from https://www.heartandstroke.ca/heart-disease/conditions/atherosclerosis)

Explain how a build-up of a fatty deposit in an artery carrying blood to the heart muscle can cause heart disease.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 7 · part a · 4 marks

June 2024 Paper 2

Worked-example retry · Attempt part a only.

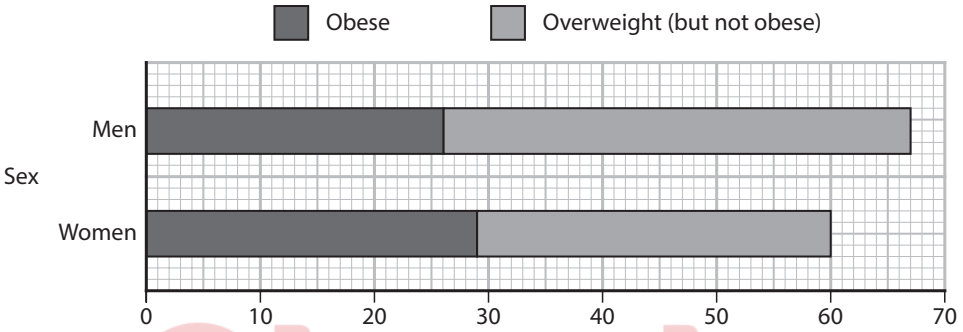
DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) People with obesity are more at risk of heart disease than those who are overweight.

The chart shows the percentage of men and women in the United Kingdom who were either overweight or obese in 2018.



(Source: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2018/summary)

(i) In 2018 there were 66.5 million people living in the UK. Of these people, 33 million were men.

Calculate how many more women than men were classified as obese.

(3)

extra number of women = ...

OLevels SCHOOL

P 7 4 6 3 0 A 0 1 9 2 4

Turn over ▶

1. Question 7 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part a.ii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.ii only.

9 Aspirin is a painkiller, but can also be prescribed to people who are at risk of having a stroke or a heart attack.

Aspirin dissolves blood clots and is also an enzyme inhibitor, which reduces the risk of more blood clots forming in blood vessels.

(a) (i) Describe how a blood clot forms in a blood vessel.

(3)

(ii) Explain how a blood clot in the coronary artery increases the risk of a heart attack.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

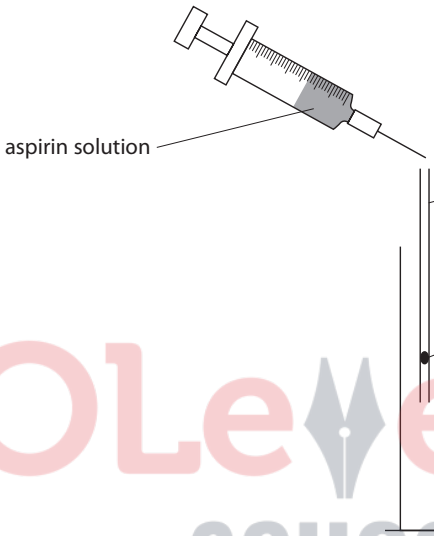
OLevels.com SCHOOL

2. Question 9 · part a.ii · 3 marks

June 2019 Paper 2

Further practice · Attempt part a.ii only.

(b) A student uses this apparatus to investigate the effect of aspirin on the rate of blood flow through blood vessels.



The student uses this method.

- dissolve one aspirin tablet in water
- use a syringe to pass the aspirin solution through a straw containing a blood clot
- record the total volume of aspirin solution in the measuring cylinder every five minutes

The student repeats the method with solutions formed from two aspirin tablets and then from three aspirin tablets.

(i) Give a control variable for this investigation.

(1)

(ii) Describe a suitable control test for this investigation.

(1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 5 8 5 6 6 A 0 2 4 2 8

2. Question 9 · part a.ii · 3 marks · Working and answer

Working and answer

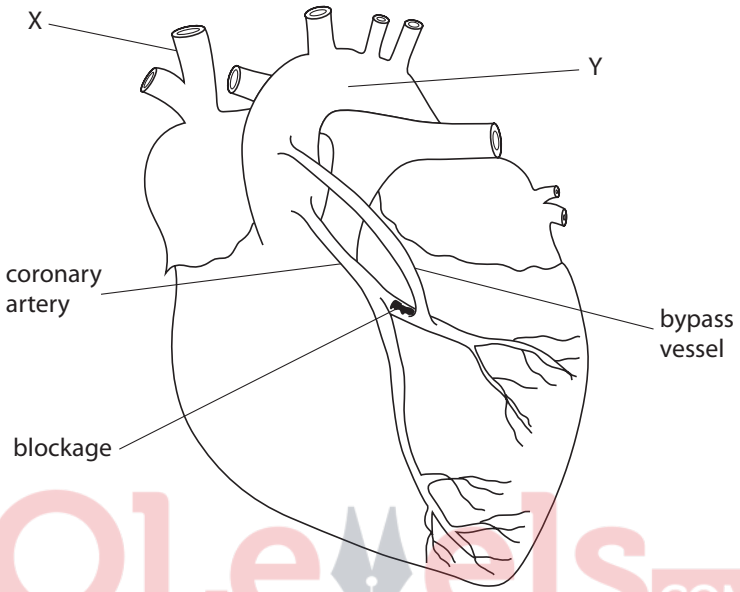
OLevels.com SCHOOL

3. Question 3 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



OLevels.com SCHOOL

(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

3. Question 3 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

OLevels.com SCHOOL

(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

3. Question 3 · part b.i · 1 mark · Working and answer

Working and answer

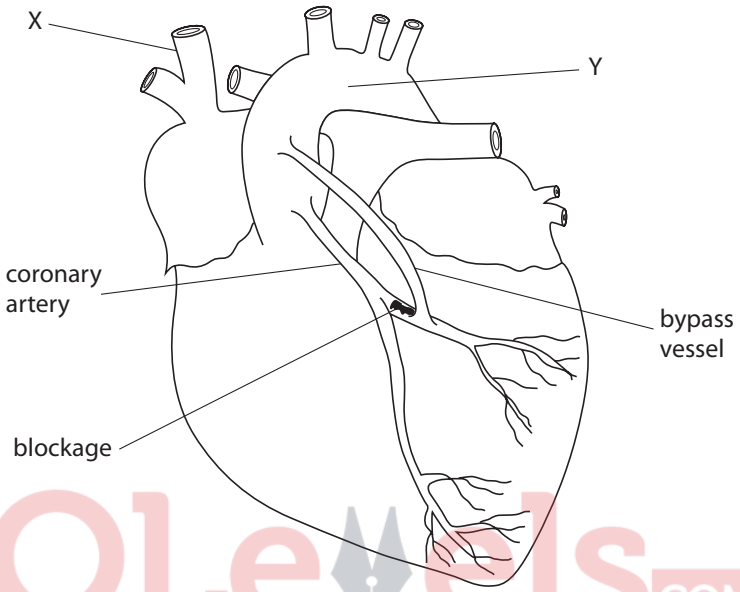
OLevels.com SCHOOL

4. Question 3 · part b.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

3 The diagram shows a person's heart. The person has had a heart bypass operation because a coronary artery had become partly blocked.



(a) (i) What is the name of blood vessel Y?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) To which part of the heart is blood vessel X connected?

(1)

- ☒ **A** left atrium
- ☒ **B** left ventricle
- ☒ **C** right atrium
- ☒ **D** right ventricle

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

4. Question 3 · part b.ii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) (i) Before the bypass operation, the coronary artery was becoming blocked at the point shown on the diagram.

Draw a circle on the diagram to show the area of the heart that was affected by the blockage before the bypass operation.

(1)

(ii) Describe what effect this blockage would have on the heart tissue if the bypass vessel was not inserted.

(3)

OLevels.com SCHOOL

(iii) Suggest how the bypass vessel allows the heart to work more effectively.

(2)

(Total for Question 3 = 8 marks)

4. Question 3 · part b.ii · 3 marks · Working and answer

Working and answer

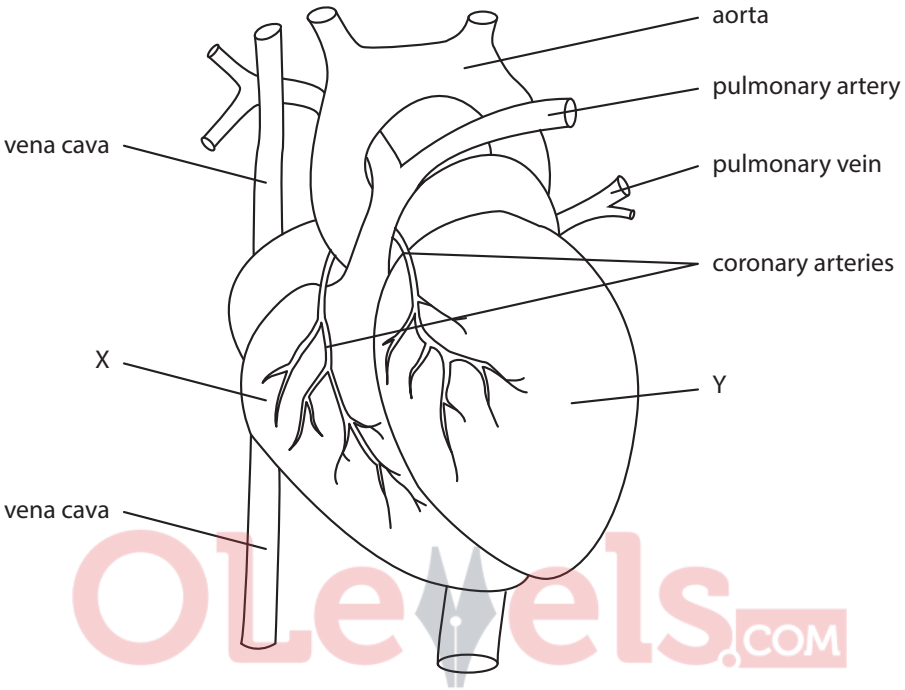
OLevels.com SCHOOL

5. Question 7 · part c · 3 marks

November 2023 Paper 1

Further practice · Attempt part c only.

7 The diagram shows the heart as seen from the front.



(a) The vena cava carries deoxygenated blood.

Name a different blood vessel, shown on the diagram, that also carries deoxygenated blood.

(1)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 7 · part c · 3 marks

November 2023 Paper 1

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Blood chambers X and Y have walls that are different in thickness.

Explain this difference.

(5)

OLevels.com SCHOOL

(c) The coronary arteries can become blocked.

Explain the effect of this blockage on the functioning of the heart.

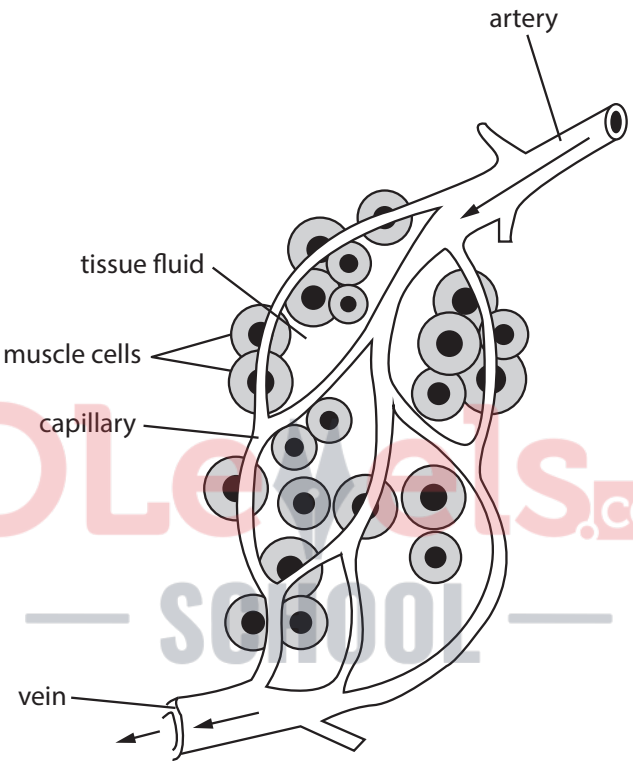
(3)

5. Question 7 · part c · 3 marks

November 2023 Paper 1

Further practice · Attempt part c only.

(d) The diagram shows a capillary bed.



Use the diagram to explain the function of the tissue fluid.

(3)

(Total for Question 7 = 12 marks)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN T

5. Question 7 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

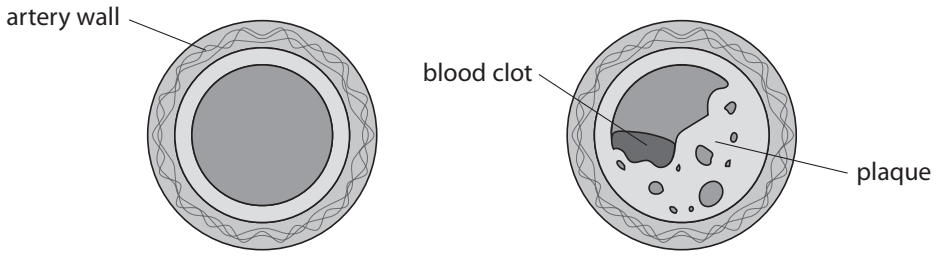
6. Question 6 · part a · 4 marks

June 2024 Paper 2R

Further practice · Attempt part a only.

6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

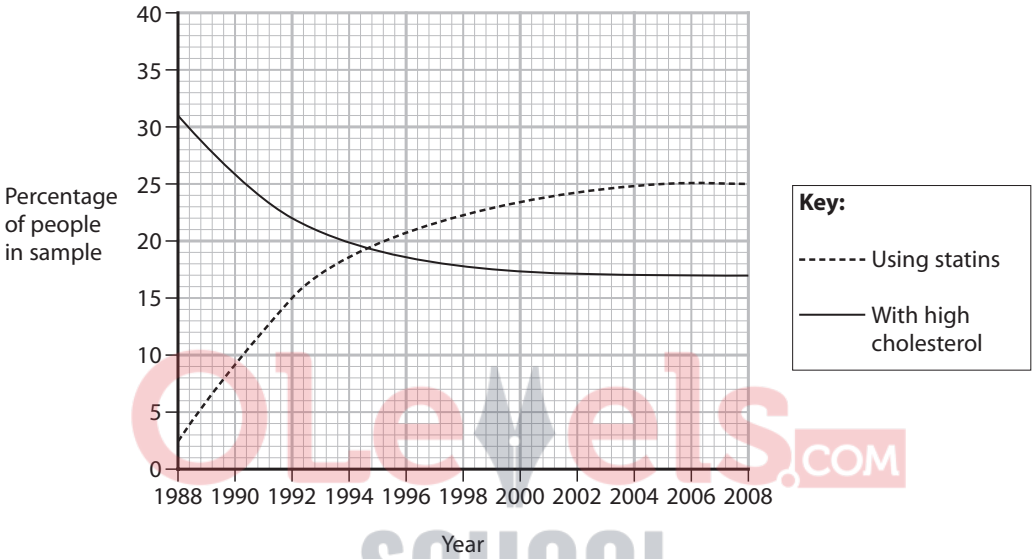
6. Question 6 · part a · 4 marks

June 2024 Paper 2R

Further practice · Attempt part a only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

.....

.....

(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 4 4 9 5 A 0 1 8 2 8

6. Question 6 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 4 · part c · 2 marks

November 2024 Paper 1

Further practice · Attempt part c only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

(2)

(ii) Name the hormone that stimulates the liver to produce glycogen.

(1)

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

(1)

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

BMI	Category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25 to 29.9	overweight
30 to 34.9	obese
35 to 39.9	severely obese
more than 39.9	morbidly obese

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

7. Question 4 · part c · 2 marks

November 2024 Paper 1

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

(i) A person has a height of 1.69 m and a mass of 67.0 kg.

Calculate the BMI of this person.

(2)

BMI .....

(ii) Give the BMI category of this person.

(1)

.....

(c) Obesity is a risk factor for heart disease.

Explain how obesity can cause heart disease.

(2)

.....

.....

.....

.....

.....

(Total for Question 4 = 9 marks)

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

Working and answer

8. Question 4 · part d · 3 marks

June 2021 Paper 2

Further practice · Attempt part d only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

OLevels.COM SCHOOL

(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

2 .....

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8

P 6 6 4 3 3 A 0 8 2 4

8. Question 4 · part d · 3 marks

June 2021 Paper 2

Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

OLevels.COM SCHOOL

(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

8. Question 4 · part d · 3 marks

June 2021 Paper 2

Further practice · Attempt part d only.

(d) Diabetes causes a narrowing of the arteries supplying blood to the lower limbs.

Suggest why this can lead to the need to amputate the lower limbs.

(3)

DO NOT WRITE IN THIS AREA

(Total for Question 4 = 14 marks)

OLevels.com SCHOOL

Working and answer